

Fine-tuning

Moving a trained model to a new task without training it over again

Required

Hu et al., LoRA: Low-Rank Adaptation of Large Language Models · [arXiv:2106.09685](#)

Optional

Dettmers et al., QLoRA · [arXiv:2305.14314](#) Sheng et al., S-LoRA · [arXiv:2311.03285](#) Zhou et al., LIMA · [arXiv:2305.11206](#)
Kulgod et al., Reducing Text Bias in Synthetic MCQAs · [arXiv:2602.17677](#)

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Quick overview

The size of adaptation. Sixteen bytes a weight, and where they go.

A 7B model needs about 120 GB to fine-tune the obvious way

What the model sees. Templates, loss masking, packing.

The prompt is context; only the answer is scored

Teachers as datasets. Distillation, and the leak it can carry.

Carried over from the data lecture, with our own failure case

Low-rank adaptation. LoRA, the rank knee, four-bit adapters.

Measured on your Assignment 1 checkpoint

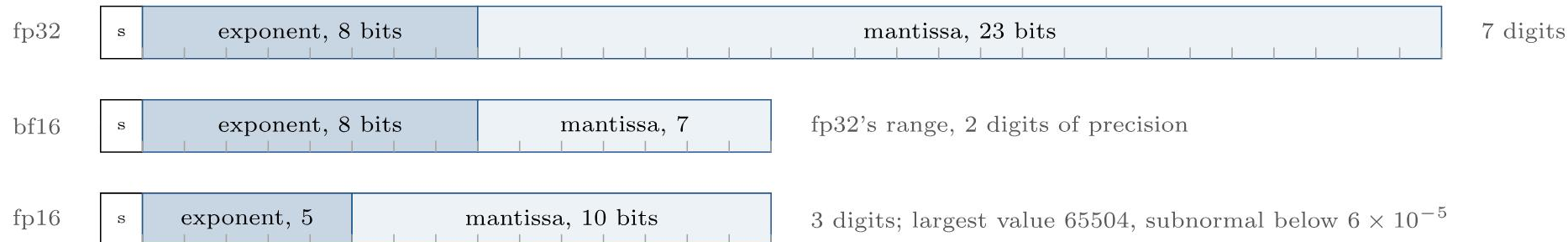
When it is not enough. Full fine-tuning, small data, the ceiling.

Where post-training picks up on October 29

1 THE SIZE OF ADAPTATION

Sixteen Bytes a Weight

Number Formats

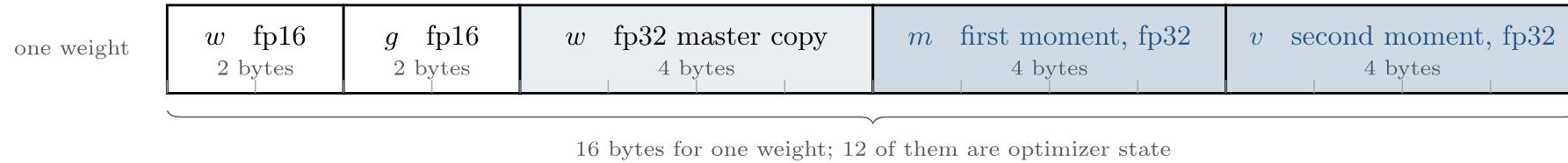


a float divides its bits into a sign, an exponent for range, and a mantissa for precision

digits: precision comes from the mantissa plus its implied leading one. 24 bits give about 7 decimal digits, 11 bits about 3, 8 bits about 2.

subnormal: below the smallest exponent the leading one is dropped and precision shrinks toward zero. fp16 enters that regime under 6×10^{-5} and reaches zero under 6×10^{-8} .

Sixteen Bytes a Weight



$$m \leftarrow \beta_1 m + (1 - \beta_1) g$$

running average of the gradient, $\beta_1 = 0.9$

$$v \leftarrow \beta_2 v + (1 - \beta_2) g^2$$

running average of the squared gradient, $\beta_2 = 0.999$

$$w \leftarrow w - \eta \hat{m} / (\sqrt{\hat{v}} + \epsilon)$$

the step, applied to the fp32 copy; hats are the bias corrections

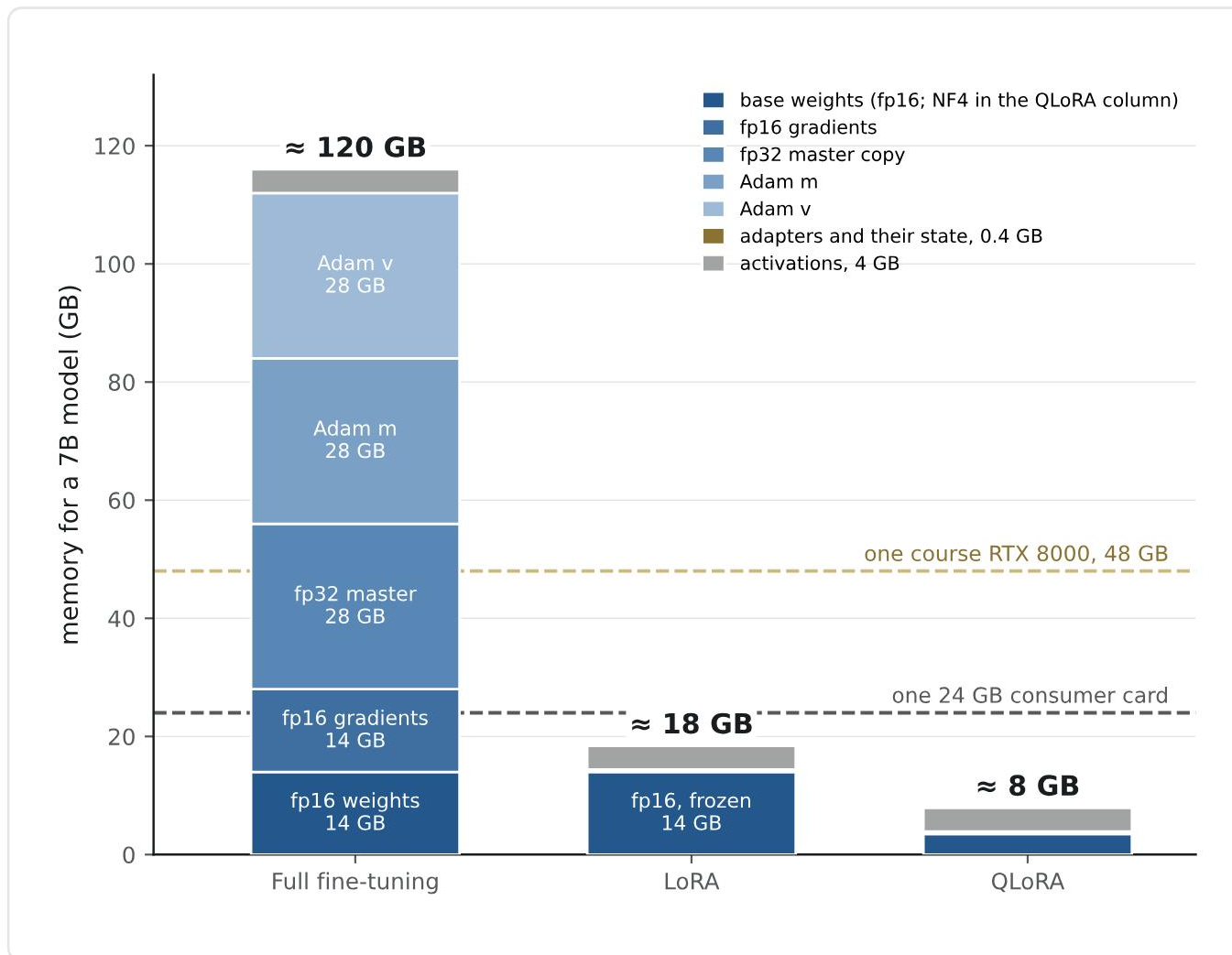
fp16 bottoms out near 6×10^{-8} ; a gradient of 10^{-4} squared is 10^{-8} , so v would read 0

first moment m : a running average of the gradient, one fp32 number per weight, kept from step to step.

second moment v : a running average of the squared gradient, one more fp32 number per weight. The step divides by its square root.

why 32 bits: the moments accumulate tiny increments over thousands of steps. fp16 cannot hold a squared gradient of 10^{-8} , and bf16 cannot resolve the increment.

Sixteen Bytes a Weight



the sixteen: 2 bytes fp16 weights, 2 gradients, 4 fp32 master copy, 4 Adam *m*, 4 Adam *v*. The weights are an eighth of the total.

what moves: freezing the base deletes the gradient, the master copy and both Adam states, 14 of the 16 bytes.

what stays: the activations, and the frozen weights themselves. Four-bit storage of the frozen copy is the last factor of two.

2

SUPERVISED FINE-TUNING

What the Model Sees

Templates and Masking

stored example

user: Write a poem

assistant: Shall I compare thee

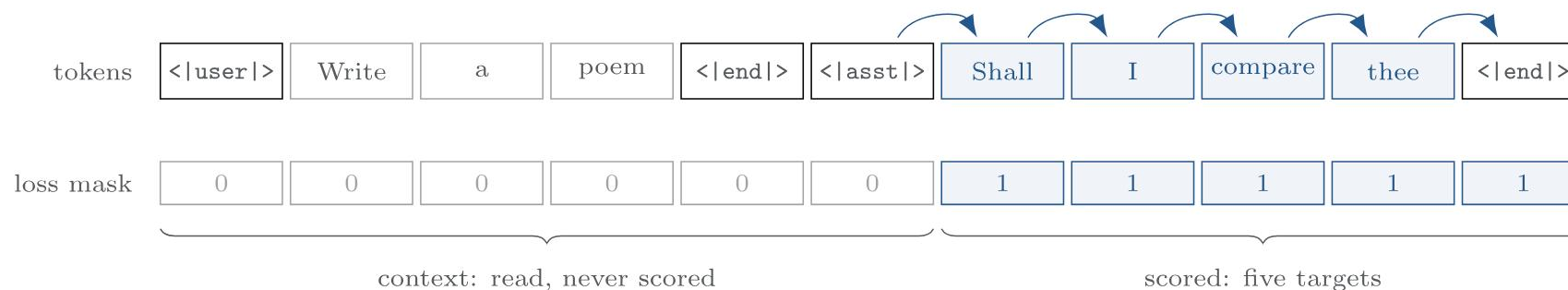
apply

chat template

<|user|> {prompt} <|end|>

<|asst|> {answer} <|end|>

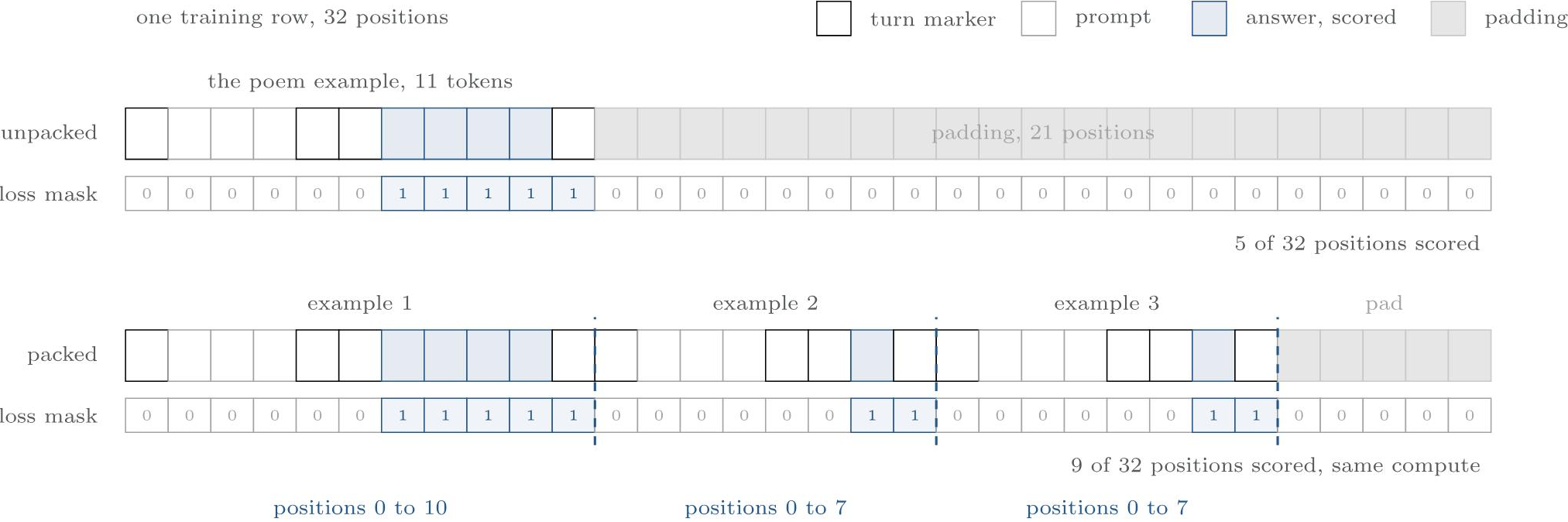
every position predicts the next token; these five predictions are scored



$$\mathcal{L} = -\frac{1}{5} \sum_{t \in \text{scored}} \log p(x_t | x_{<t})$$

the prompt is context; score it too and the model also learns to write your questions

Packing



each example keeps its own positions, and attention does not cross the dashed lines

Schematic. 32-position rows drawn; Assignment 1 trains on 256-token rows, where this 11-token example alone would leave 245 positions as padding.

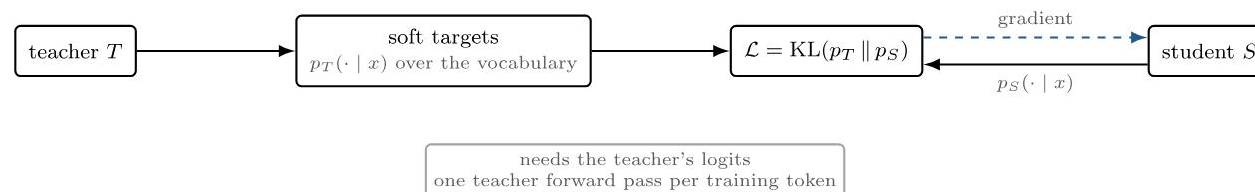
3 DISTILLATION

Teachers as Datasets

Two Ways to Distill

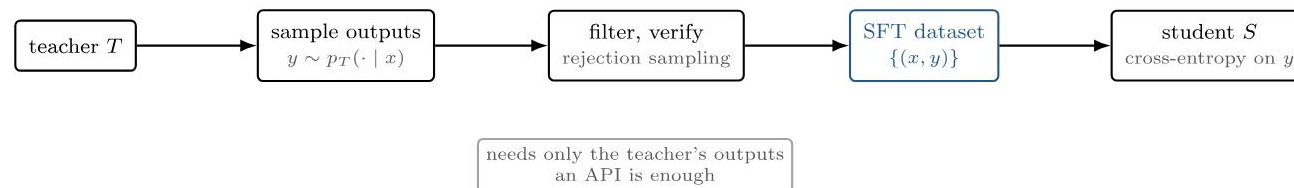
logit distillation

Hinton, Vinyals and Dean, 2015



data distillation

no logits needed: this is how it is done at scale



$$\mathcal{L}_{\text{logit}} = \text{KL}(p_T(\cdot | x) \| p_S(\cdot | x))$$

$$\mathcal{L}_{\text{data}} = -\log p_S(y | x), \quad y \sim \text{filter}(p_T(\cdot | x))$$

logit distillation: the soft targets carry the teacher's whole ranking of the vocabulary, so a near-miss token still teaches. Practical when you own the teacher.

data distillation: sample, verify, keep; the filter is the curation step. The route when the teacher is behind an API, which is every frontier model.

R1 Distills

800k samples written by DeepSeek-R1: 600k reasoning traces kept by rejection sampling, 200k other; then plain SFT on Qwen2.5 and Llama-3 bases, no RL on the students

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

GPT-4o: 9.3

a distilled 7B: 55.5

the 32B: 72.6

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCodeBench
	pass@1	cons@64	pass@1	pass@1	pass@1
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6	55.0	40.2
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2

RL on the base: 47.0.
Distilling into it:
72.6.

Zero Vision

Zoox. Qwen2-VL-2B fine-tuned on multiple-choice questions that Gemini 2.5 wrote from ground-truth maneuver labels; then tested with the video zeroed out

Dataset	Training Strategy	Accuracy (%) ↑		Accuracy Video Zeroed Out (%)		
		Normal	Shuffled	(D_{NV}) ↑	(D_N) ↓	(D_V) ↓
D_{llm}	No Training	29.23%	6.46%	17.48%	32.22%	23.4% (-1.6%)
D_{llm}	Regular SFT	93.82%	89.03%	99.68%	64.66%	91.9% (+66.9%)
D_{new}	No Training	24.38%	6.23%	12.94%	29.85%	23.1% (-1.9%)
D_{new}	Regular SFT	75.72%	66.43%	100%	0.21%	27.9% (+2.9%)
D_{new}	Curriculum	77.28%	70.82%	100%	0.84%	30.4% (+5.4%)
D_{new}	Regular SFT (FF)	88.08%	84.35%	100%	0.21%	29.5% (+4.5%)
D_{new}	Curriculum (FF)	92.09%	87.64%	100%	0.0%	29.9% (+4.9%)

93.8% with the video

91.9% without it: 66.9 points above random

rows and columns: D_{llm} has generator-written distractors, D_{new} is the decoupled set. D_{NV} : "agent not visible" is the answer; D_N : present but wrong; D_V : absent. Random is 25%.

no training: Gemini 2.5 Pro, text only, scores 38.1% on the same questions

what people saw: manual inspection found no giveaway in length, wording or position

Decoupling

Our Solution

distractors: other scenes' true answers, agent id rewritten

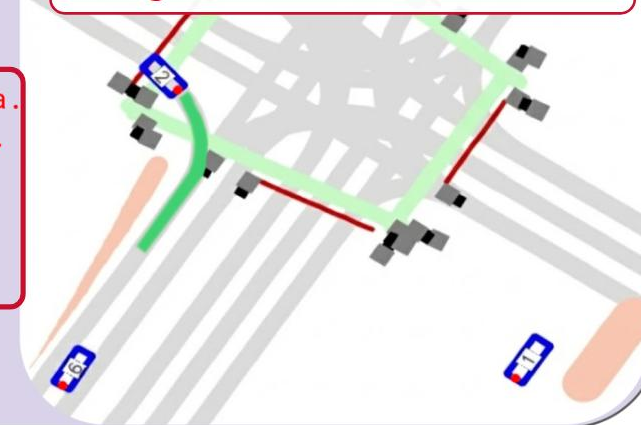
What maneuver is agent 2 executing in this scenario?



Pool of human-verified maneuvers & LLM descriptions

- A:Agent 2 is getting on the road from an off-road area.
- B:Cannot determine maneuver as agent 2 is not visible.
- C:Agent 2 is proceeding straight at the traffic light junction.
- Heuristic Distractor Selection
- D:Agent 2 is initially stopped at a traffic light junction and then proceeds to turn.

the same generator wrote every line; the wrong ones no longer point at the right one



Our method decouples text from visual grounding. Results show bias exploitation drops to near-random levels (-5.01% Zero-Shot / +2.88% SFT)

curriculum: drop the options early in SFT so the model answers from the video; bring them back gradually

video off: from 66.9 points above random to 2.9 with plain SFT, 4.9 with the curriculum and full fine-tuning

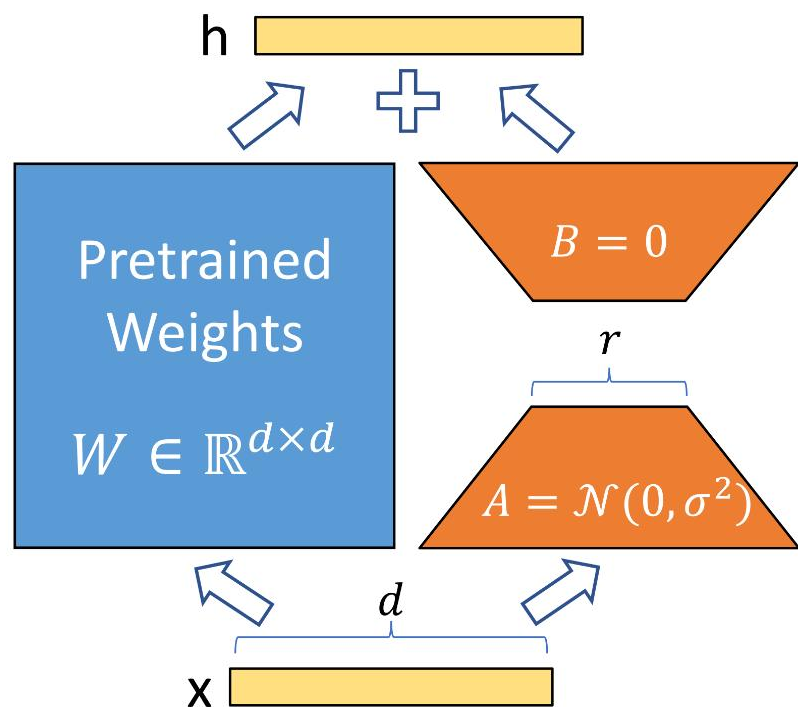
video on: 92.1% with the curriculum and full fine-tuning; D_N falls from 65% to zero

4

LOW-RANK ADAPTATION

A Smaller Update

Low-Rank Update



$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x$$

$W_0 \in \mathbb{R}^{d \times k}$ frozen. $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$, with $r \ll \min(d, k)$ chosen by you.

$B = 0$ at initialization, so $\Delta W = 0$ and the adapted model starts as the base model, unchanged.

At inference, fold $\frac{\alpha}{r} B A$ into W_0 ; the served model keeps the base model's shape and latency.

intrinsic rank: the update a task needs has low rank even when W_0 has full rank. That is the paper's hypothesis, and Section 7 of the paper tests it.

α/r : a scaling convention. Hold α proportional to r and the same learning rate transfers across ranks.

Low-Rank Update

One attention projection, $d = k = 4096$:

W_0 : $4096 \times 4096 \approx 17\text{M}$, frozen

A and B at $r = 16$: $2 \times 16 \times 4096 \approx 130\text{k}$ trainable

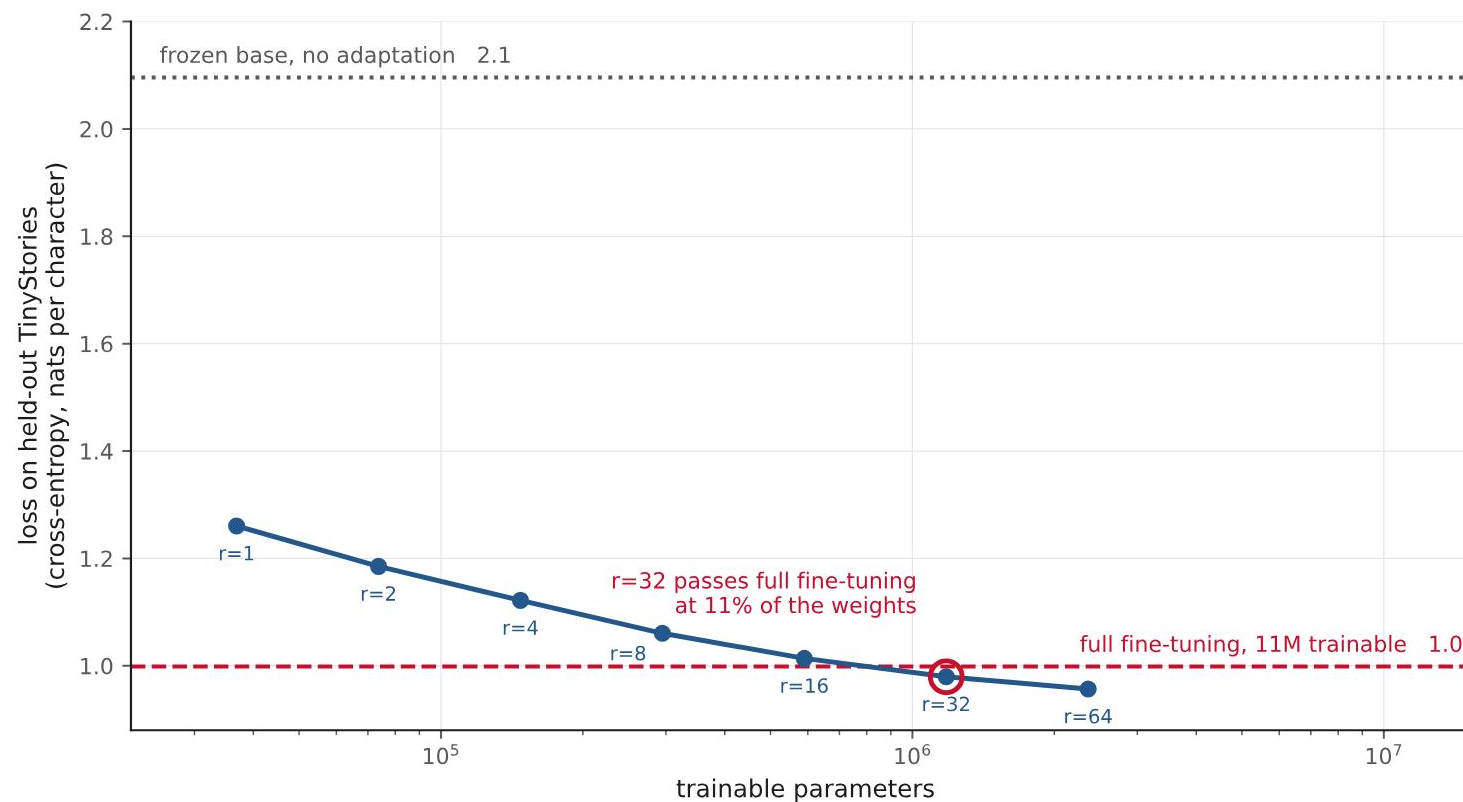
BA is 4096×4096 , the shape of W_0 : computed, never stored.

128 times fewer, and only these get optimizer state.

On our Assignment 1 base model, $d = 384$ and $r = 8$ on the qkv, output and MLP projections gives about 290k trainable parameters, 2.8% of the model.

Model&Method	# Trainable Parameters	WikiSQL	MNLI-m	SAMSum
		Acc. (%)	Acc. (%)	R1/R2/RL
GPT-3 (FT)	175,255.8M	73.8	89.5	52.0/28.0/44.5
GPT-3 (BitFit)	14.2M	71.3	91.0	51.3/27.4/43.5
GPT-3 (PreEmbed)	3.2M	63.1	88.6	48.3/24.2/40.5
GPT-3 (PreLayer)	20.2M	70.1	89.5	50.8/27.3/43.5
GPT-3 (Adapter ^H)	7.1M	71.9	89.8	53.0/28.9/44.8
GPT-3 (Adapter ^H)	40.1M	73.2	91.5	53.2/29.0/45.1
GPT-3 (LoRA)	4.7M	73.4	91.7	53.8/29.8/45.9
GPT-3 (LoRA)	37.7M	74.0	91.6	53.4/29.2/45.1

The Rank Knee



the run: Assignment 1 base checkpoint, Shakespeare-trained, adapted to the Assignment 2 TinyStories split. 600 steps, batch 32, LoRA on the qkv, output and MLP projections, alpha = 2r, one seed.

Our own runs on the Assignment 1 codebase: the Shakespeare base checkpoint adapted to the Assignment 2 TinyStories split, one seed. Full fine-tuning at the same 600 steps, learning rate $1e-4$ against $1e-3$ for the LoRA arms.

Which Matrices

	# of Trainable Parameters = 18M						
Weight Type Rank r	W_q 8	W_k 8	W_v 8	W_o 8	W_q, W_k 4	W_q, W_v 4	W_q, W_k, W_v, W_o 2
WikiSQL ($\pm 0.5\%$)	70.4	70.0	73.0	73.2	71.4	73.7	73.7
MultiNLI ($\pm 0.1\%$)	91.0	90.8	91.0	91.3	91.3	91.3	91.7

one matrix, rank 8: 70.4
on WikiSQL

four matrices, rank 2,
same 18M parameters:
73.7

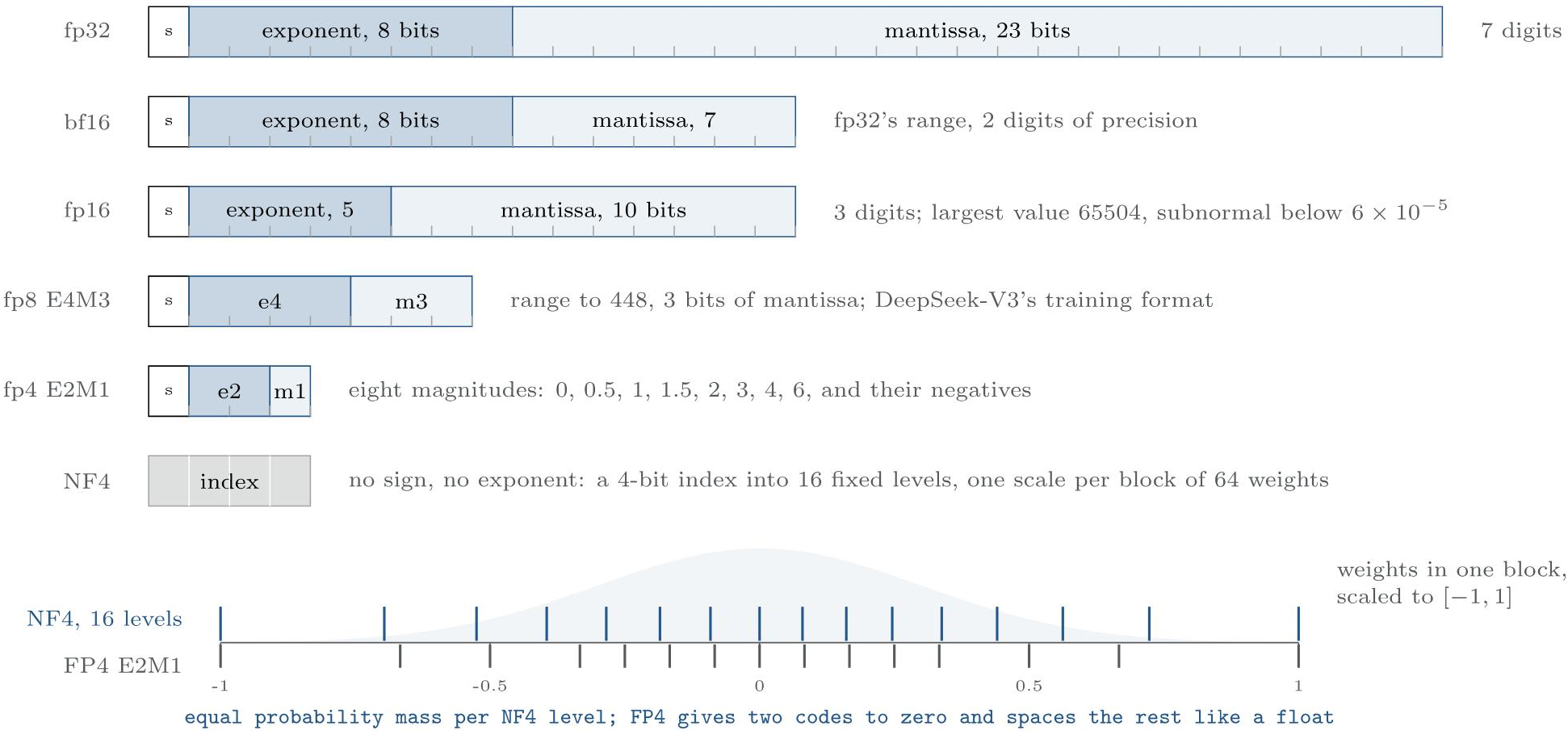
	Weight Type	$r = 1$	$r = 2$	$r = 4$	$r = 8$	$r = 64$
WikiSQL($\pm 0.5\%$)	W_q	68.8	69.6	70.5	70.4	70.0
	W_q, W_v	73.4	73.3	73.7	73.8	73.5
	W_q, W_k, W_v, W_o	74.1	73.7	74.0	74.0	73.9
MultiNLI ($\pm 0.1\%$)	W_q	90.7	90.9	91.1	90.7	90.7
	W_q, W_v	91.3	91.4	91.3	91.6	91.4
	W_q, W_k, W_v, W_o	91.2	91.7	91.7	91.5	91.4

W_q, W_v at rank 1: 73.4;
at rank 64: 73.5, inside
the 0.5 noise

5 FOUR-BIT ADAPTERS

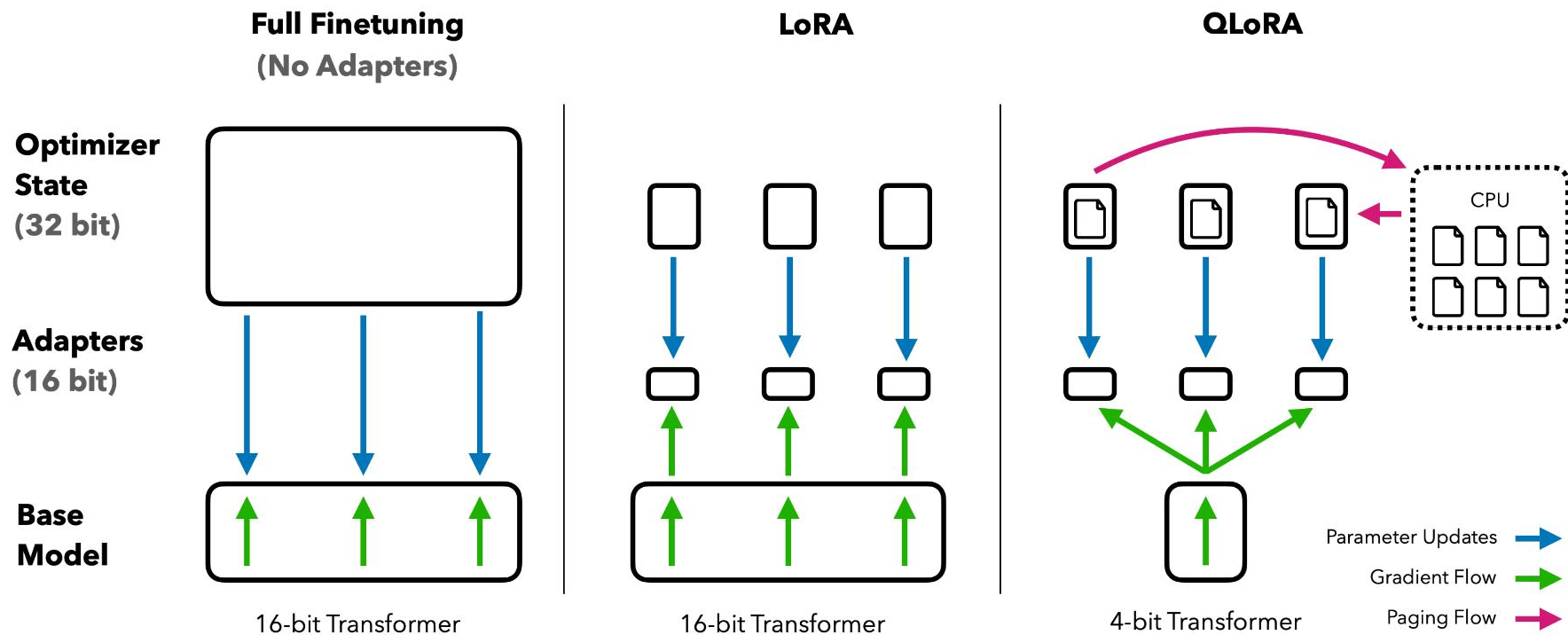
One Card

Number Formats



The three floats from section one; fp8 E4M3 and fp4 E2M1 after the OCP microscaling formats. NF4 from Dettmers et al., "QLoRA," 2023, arXiv:2305.14314, Section 3, with the sixteen levels listed in its appendix. Number formats in full on October 13.

Four Bits



NF4: a 4-bit type whose 16 levels are the quantiles of a normal distribution, which is what a trained weight matrix looks like.

double quantization: the per-block scale factors are themselves quantized, saving about 0.37 bits a parameter.

paged optimizers: optimizer state moves to CPU memory on a spike, so a long sequence does not end the run.

Four Bits

LLaMA Size Dataset	Mean 5-shot MMLU Accuracy								Mean
	7B		13B		33B		65B		
	Alpaca	FLAN v2	Alpaca	FLAN v2	Alpaca	FLAN v2	Alpaca	FLAN v2	
BFloat16	38.4	45.6	47.2	50.6	57.7	60.5	61.8	62.5	53.0
Float4	37.2	44.0	47.3	50.0	55.9	58.5	61.3	63.3	52.2
NFloat4 + DQ	39.0	44.5	47.5	50.7	57.3	59.2	61.8	63.9	53.1

BFloat16 mean: 53.0

NFloat4 + DQ mean: 53.1, within noise of 16-bit

Float4 mean: 52.2, nearly a point down

reading the table: four model sizes, two finetuning sets, 5-shot MMLU. The Mean column averages all eight.

the comparison: same base, same data, same recipe. Only the storage type of the frozen weights changes.

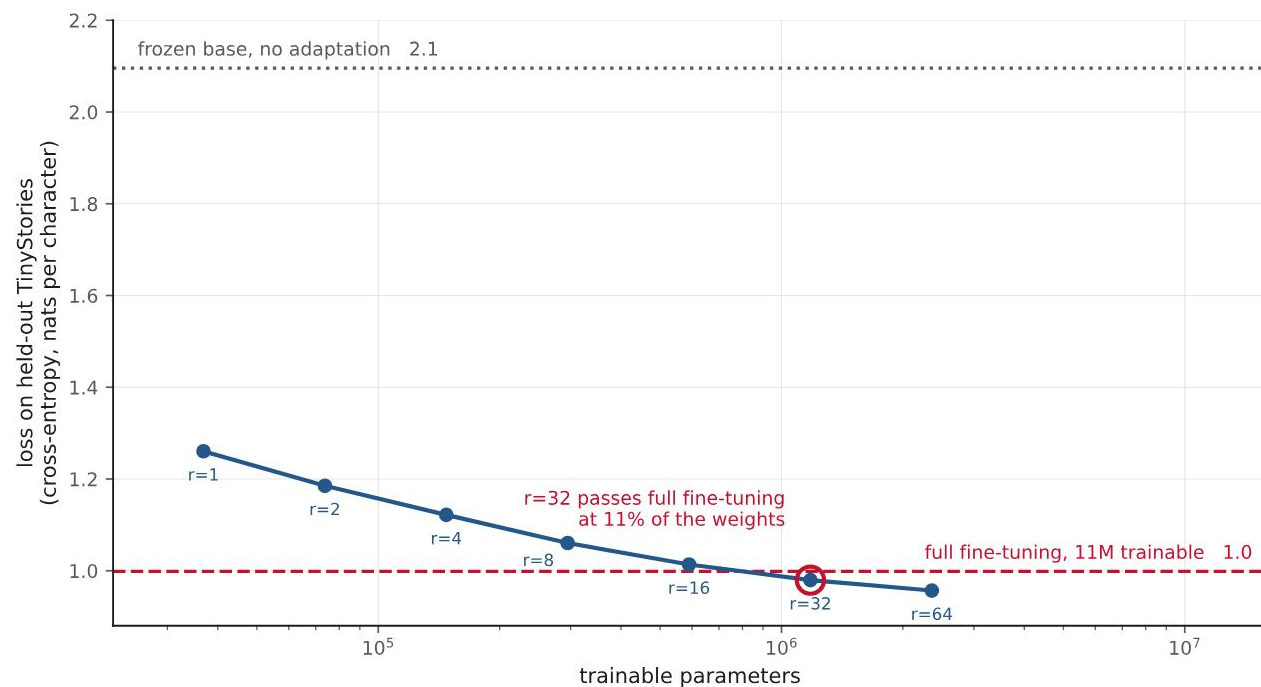
the caveat: this finetunes a quantized base. Quantizing an already-finetuned model is October 13.

6

WHERE THE SMALL UPDATE STOPS

Not Always Enough

The Counterpoint



Small models. Around 2B, full fine-tuning fits on one 48 GB card, and the memory argument is gone.

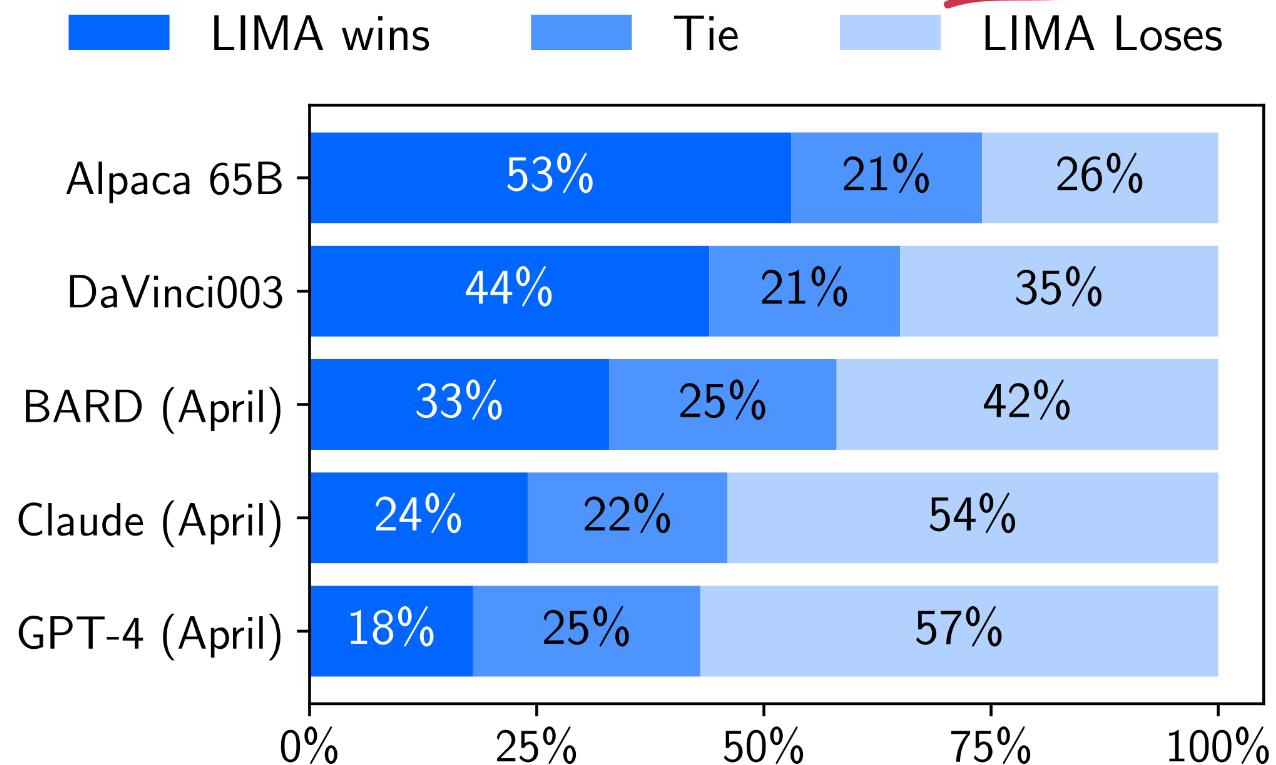
Large shifts. A new language or modality needs more capacity than a low-rank update carries.

Forgetting. Full fine-tuning overwrites what the base knew; replaying a slice of the original mixture is the standard guard.

Our own run measured a narrow claim: matched steps, one seed, an 11M model, and a full arm that may be undertrained.

Less Data

vs Human

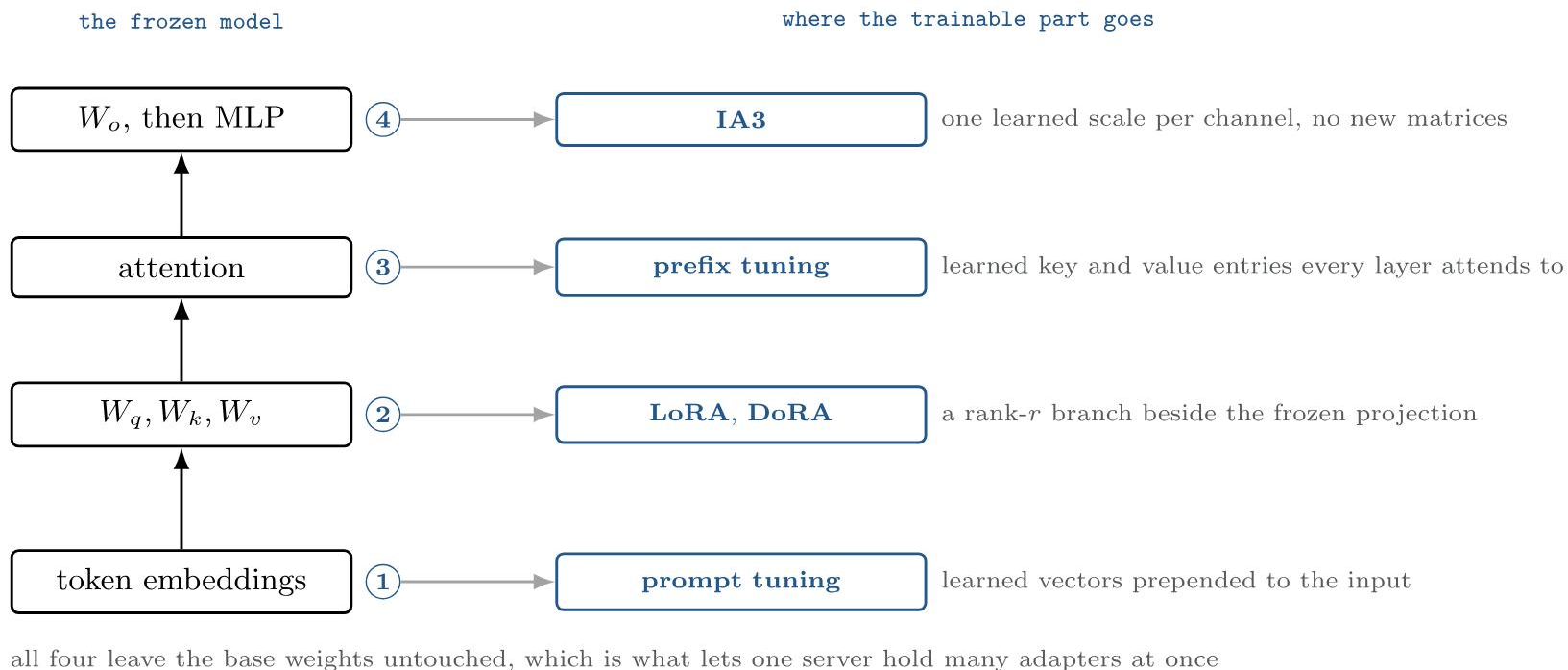


LIMA: a 65B base, 1,000 curated examples, plain SFT and no RLHF.

Preferred to Alpaca 65B, which was fine-tuned on 52,000 generated examples.

Zhou et al. call it the superficial alignment hypothesis: pretraining supplies the knowledge, instruction tuning selects the format, so a small, clean, diverse set is enough.

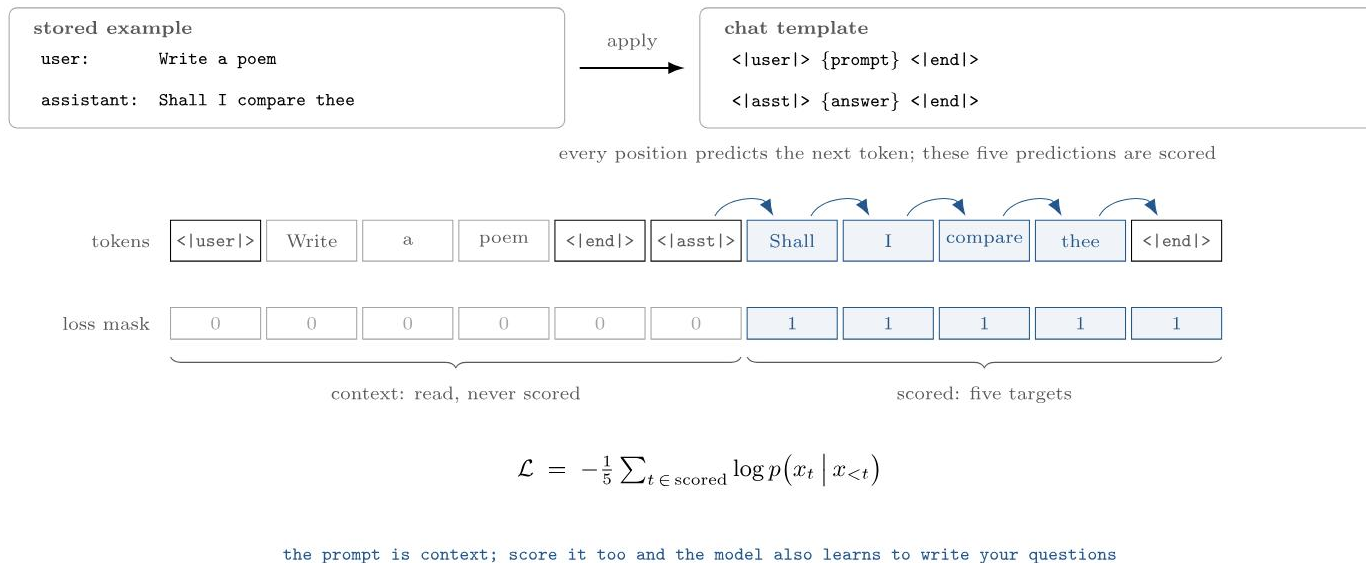
Beyond LoRA



at inference: prompt and prefix tuning lengthen every sequence; LoRA, DoRA and IA3 fold into the weights and add no latency.

serving them: batching and paging adapters across requests is S-LoRA, and it belongs to Mark's inference systems lectures, October 15 to 22.

The Ceiling



Mode averaging. Two acceptable answers to one prompt, and cross-entropy pushes toward the average of both.

Compounding. It trained on correct prefixes and generates from its own; one wrong token puts it in a state it never saw.

No preference. The loss knows matching and not matching; nothing in it says one answer is better than another.

Reading

Required

Hu et al., LoRA: Low-Rank Adaptation of Large Language Models · [arXiv:2106.09685](#)

Optional

Dettmers et al., QLoRA · [arXiv:2305.14314](#) Sheng et al., S-LoRA · [arXiv:2311.03285](#) Zhou et al., LIMA · [arXiv:2305.11206](#)

Kulgod et al., Reducing Text Bias in Synthetic MCQAs · [arXiv:2602.17677](#)

Assignment 5, out Tuesday: LoRA on a real checkpoint, your own rank sweep, and the memory model from section one checked against what the GPU reports.